

THE EMERGENCE OF MATTER

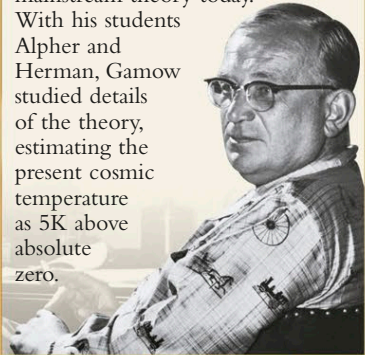
About 1 microsecond (10^{-6} or one millionth of a second) after the Big Bang, the young universe contained, in addition to vast quantities of radiant energy, or photons, a seething “soup” of quarks, antiquarks, and gluons. Also present were the class of elementary particles called leptons (mainly electrons, neutrinos, and their antiparticles) forming from energy and then being annihilated back to energy. The stage was set for the next processes of matter formation that led to our current universe. First, quarks and gluons met to make heavier particles—particularly protons and a smaller number of neutrons. Next, the neutrons combined with some of the protons to form atomic nuclei, mainly those of helium. The remaining protons, destined to form the nuclei of hydrogen atoms, stayed uncombined. Finally, after about 400,000 years, the universe cooled sufficiently for electrons to combine with the free protons and helium nuclei—so forming the first atoms.

THE NEXT HALF-MILLION YEARS

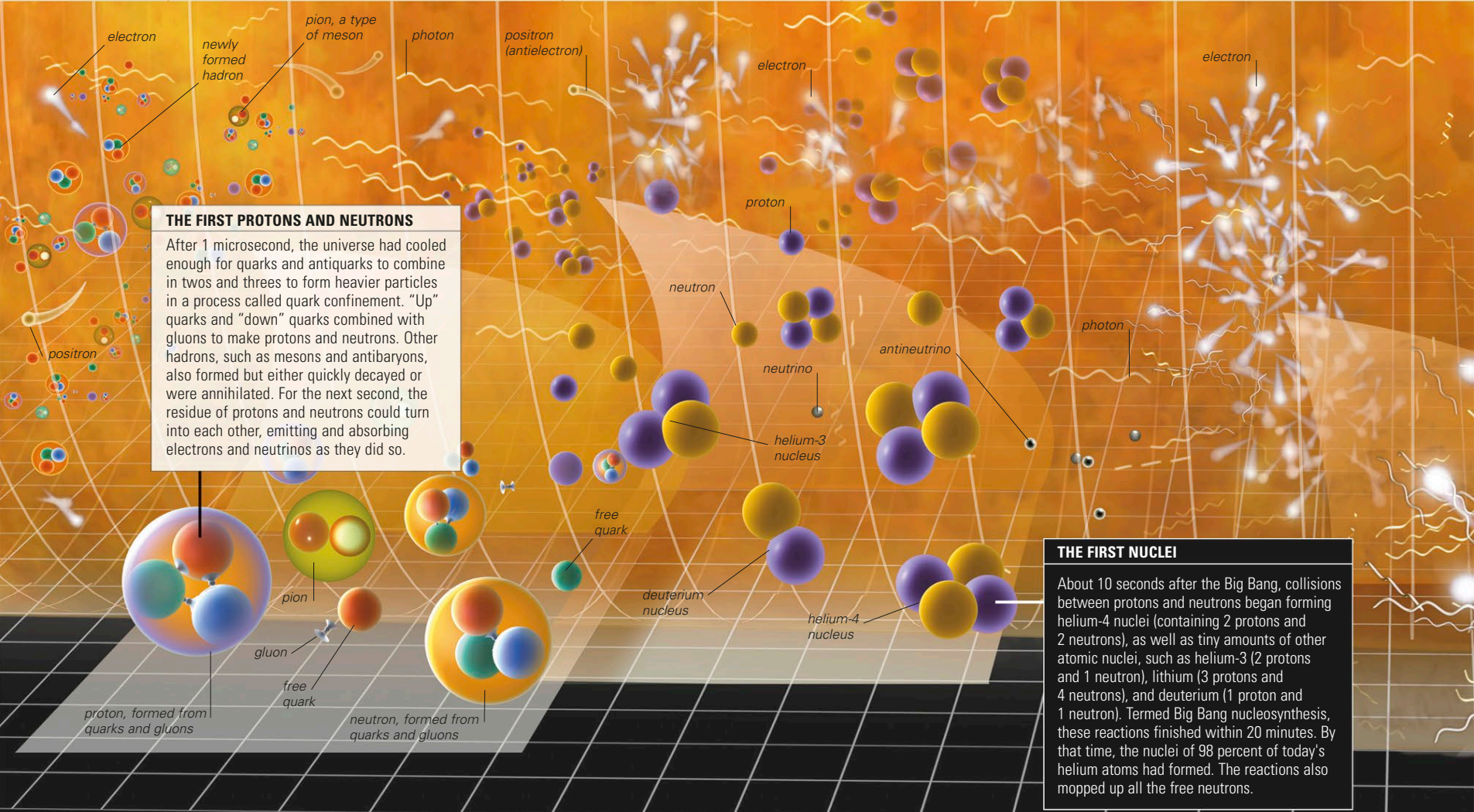
The timeline on these two pages shows events from 1 microsecond to about 500,000 years after the Big Bang. The temperature dropped from 10^{11}°K (180 billion $^{\circ}\text{F}$ /100 billion $^{\circ}\text{C}$) to $4,500^{\circ}\text{F}$ ($2,500^{\circ}\text{C}$). Today's observable universe expanded from about 50 light-hours (100 billion km) to many millions of light-years wide.

GEORGE GAMOW

Influenced by the original “Big Bang” concept of Georges Lemaître, Ukrainian American physicist George Gamow (1904–1968) played a major role in developing the “hot Big Bang” theory. This, supplemented by inflation, is the mainstream theory today. With his students Alpher and Herman, Gamow studied details of the theory, estimating the present cosmic temperature as 5K above absolute zero.



DIAMETER	60 billion miles/100 billion km	10 light-years (1 light-year = 5.88 million miles/9.46 trillion km)	100 light-years
TEMPERATURE	10^{11}K (180 billion $^{\circ}\text{F}$ /100 billion $^{\circ}\text{C}$)	10^{10}K (18 billion $^{\circ}\text{F}$ /10 billion $^{\circ}\text{C}$)	10^9K (1.8 billion $^{\circ}\text{F}$ /1 billion $^{\circ}\text{C}$)
	HADRON EPOCH Around the beginning of this epoch, quarks and antiquarks began combining to form particles called hadrons. These included baryons (protons and neutrons), antibaryons, and mesons.	LEPTON EPOCH During this epoch, leptons (electrons, neutrinos, and their antiparticles) were very numerous. By its end, the electrons annihilated with positrons (antielectrons).	BIG BANG NUCLEOSYNTHESIS Neutrons gradually converted into protons as the universe cooled, but when there was about one neutron for every seven protons, most remaining neutrons combined with protons to make helium nuclei, each with two protons and two neutrons.
TIME	1 microsecond 10^{-6} seconds—1 millionth of a second	1 second	10 seconds



THE FIRST PROTONS AND NEUTRONS

After 1 microsecond, the universe had cooled enough for quarks and antiquarks to combine in twos and threes to form heavier particles in a process called quark confinement. “Up” quarks and “down” quarks combined with gluons to make protons and neutrons. Other hadrons, such as mesons and antibaryons, also formed but either quickly decayed or were annihilated. For the next second, the residue of protons and neutrons could turn into each other, emitting and absorbing electrons and neutrinos as they did so.

THE FIRST NUCLEI

About 10 seconds after the Big Bang, collisions between protons and neutrons began forming helium-4 nuclei (containing 2 protons and 2 neutrons), as well as tiny amounts of other atomic nuclei, such as helium-3 (2 protons and 1 neutron), lithium (3 protons and 4 neutrons), and deuterium (1 proton and 1 neutron). Termed Big Bang nucleosynthesis, these reactions finished within 20 minutes. By that time, the nuclei of 98 percent of today's helium atoms had formed. The reactions also mopped up all the free neutrons.